UNIVERSITY OF KARACHI

COMPUTER SCIENCE Dept.

REQUIREMENT FOR FYP - I

SOFTWARE REQUIREMENT SPECIFICATION

Project Title: App name

SOFTWARE REQUIREMENT SPECIFICATION

SUBMITTED TO: Ms. Maryam Feroze

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1. Introduction

Court Connect is a web-based sports venue booking and management platform designed to bring players, venue owners, and tournament organizers onto a single system. The platform allows players to discover sports venues near their location, book courts for a specific timeslot, invite friends to split the payment of a booking, and go live to share their game with friends in a manner similar to popular social media platforms.

Currently, booking a court or organizing a casual match with friends is a fragmented process handled through phone calls, WhatsApp groups, and walk-in visits to venues. This makes it difficult to know real-time court availability, coordinate payments among a group of players, or discover organized tournaments happening nearby. Court Connect solves this by centralizing venue discovery, booking, payment, and tournament management into one platform.

The system supports two primary user roles:

- **Players** can register, browse venues, book courts individually or with friends via split payment, go live during their games, and take part in or create tournaments.
- **Venue Owners** can register their venue (subject to manual verification), manage their courts and available timeslots, view incoming bookings, and create tournament packages that players can request.

2. Objectives of the Project

The main objectives of Court Connect are:

- **Centralized Venue Discovery:** Display all registered venues, filtered by location, so players do not have to rely on scattered word-of-mouth information.
- **Verified Venue Onboarding:** Allow business owners to register their venue through a request form that is manually verified before the venue appears live on the platform, ensuring listing quality and authenticity.
- **Court & Timeslot Booking:** Let players view real-time availability of courts and timeslots and book directly through the website.
- **Player Matching & Split Payment:** Allow a player to invite friends to a booking and split the cost automatically, with each invited player required to accept before the booking is finalized.

- **Automatic Booking Expiry:** Protect venue owners from indefinitely held timeslots by automatically releasing a booking if payment is not completed within a fixed time window.
- **Go Live Social Feature:** Let players broadcast a live stream of their game, visible only to their friends, similar to a social media “story” feature.
- **Tournament Management:** Allow venue owners to publish tournament packages and allow players to either book an available package or submit a custom request for proposal (RFP) for a tournament.
- **Gamified Tournaments:** Let tournament organizers create custom teams, track live scores, and reward the winning team with experience points (XP) and leaderboard recognition.
- **Venue Owner Dashboard:** Provide venue owners a dashboard to manage bookings, add or edit venues and courts, and deactivate listings without deleting historical data.

3. Requirement Analysis

Requirements were derived directly from the process flows designed for Court Connect, covering the authentication flow, the player booking flow (with split payment), the venue owner dashboard, the go-live social feature, and the tournament section.

3.1 Functional Requirements

#	Requirement	Description
1	User Registration & Login	Users sign up or log in, then choose to continue as a Player or submit a request to register as a Venue Owner.
2	Venue Owner Verification	A venue-owner request form (company details) is submitted and manually verified by an admin. Approved requests unlock the Venue Owner dashboard; rejected requests return the user to the landing page.
3	Player Profile Setup	After authentication, a player sets up a profile (bio, profile picture) or chooses “Skip for now” and completes it later.

#	Requirement	Description
4	Browse Venues by Location	Players view a list of venues filtered according to their location, then drill down into a specific venue's available courts.
5	View Court & Timeslot Details	Selecting a court shows its available timeslots, including how many players are allowed and current pricing.
6	Book a Court (Solo)	A player can fill a request-to-play form (timeslot, number of players, preferred sport) to request a booking directly.
7	Match with Players & Split Payment	A player can open a dropdown of other players already booked for a timeslot, or select friends from a friends-only dropdown, to split the cost of a booking.
8	Booking Invitations	Invited players receive a notification/email to accept or decline. If any player declines, the request is cancelled and all participants are notified; the requester may still book solo.
9	Payment & Checkout	Once all invited players accept, the system proceeds to a checkout page with the individual amount due. A countdown timer auto-expires the booking if payment is not completed in time.
10	Payment Verification	The system verifies payment success. A successful payment confirms the booking and returns the player to their layout; a failed payment shows an error popup and returns to the venue details page.

#	Requirement	Description
11	Cancel Booking Request	A requester or invited player can cancel their participation or the entire request before all invitees accept; all affected players are notified.
12	Go Live	An authenticated player can start a live stream from their profile. The stream is visible only to that player's friends, and a notification is pushed to all of them.
13	View Friend's Live Stream	If a searched profile is a friend and currently live, an option to view the live stream is shown on their profile.
14	Venue Management (Owner)	A venue owner can add a new venue (with courts, timeslots, players allowed per slot, cost, and location), edit an existing venue, or deactivate a venue so it no longer appears to players.
15	Court Editing Restriction	A venue owner cannot edit a court's timeslot if a booking already exists on it; the system shows a pop-up and blocks the edit.
16	View Bookings (Owner)	A venue owner can view all bookings made against their venues and courts from a dedicated bookings page.
17	Create Tournament Package	A venue owner defines a tournament package: designated time slots, number of courts included, rules (referee, equipment, refreshments), and price. Saved packages appear as an “Available Tournament Slot.”
18	Browse & Book Tournament Packages	Players can browse available tournament packages for a venue, view package and court details, and book a listed package directly, proceeding to checkout.

#	Requirement	Description
19	Tournament Request for Proposal (RFP)	Alternatively, a player can submit a custom tournament request specifying date range, duration, expected players/teams, and maximum budget. Matching venue owners in that city receive a push notification.
20	Review & Counter Tournament Requests	A venue owner can view incoming tournament requests and either accept or counter-offer; the player is notified and can confirm and proceed to payment.
21	Create & Manage Tournament	Once a tournament booking is confirmed, the organizing player can create custom teams, invite participants from the app, and manage the tournament page.
22	Live Scoring & Leaderboard	During the tournament, participants can update their team's score. The system tracks a leaderboard and awards gamified experience points (XP) to the winning team.

3.2 Non-Functional Requirements

#	Requirement	Description
1	Performance	Venue listings and timeslot availability must load within 3 seconds under normal load. Database indexes should be used on location, venue_id, and timeslot columns.
2	Security	Passwords must be hashed before storage. Payment and split-payment data must be handled over secure (HTTPS) connections. Venue owner routes are protected and only accessible after admin verification.
3	Usability	The interface must clearly show booking status (pending, accepted, confirmed), countdown timers for

#	Requirement	Description
		payment/cancellation windows, and toast notifications for accepts, declines, and payment results.
4	Reliability	Booking transactions (invite -> accept -> pay -> confirm) must either fully complete or roll back — for example, an expired unpaid booking must reliably release its timeslot back to availability.
5	Scalability	The system should support a growing number of venues, concurrent bookings, and simultaneous live streams without performance degradation.
6	Availability	Deactivating a venue must immediately stop it from appearing in the player-facing layout, while preserving its historical booking data.
7	Compatibility	The website should work across modern browsers (Chrome, Firefox, Edge, Safari) and be responsive across desktop, tablet, and mobile devices.
8	Notification Delivery	Email and in-app/push notifications (booking invites, tournament requests, go-live alerts) must be delivered reliably and promptly.

4. System Design

This section presents the process flows (represented as Data Flow / Process Diagrams) that were designed for each module of the system, followed by the Use Case Diagram, Class Diagram, and Entity-Relationship Diagram (ERD) derived from them.

4.1 Data Flow / Process Diagrams

The following diagrams show the process flow designed for each major module of Court Connect prior to implementation.

4.1.1 Authentication Flow

Covers sign up/login, the branching between registering a venue and continuing as a player, manual verification of venue owner requests, and player profile setup.

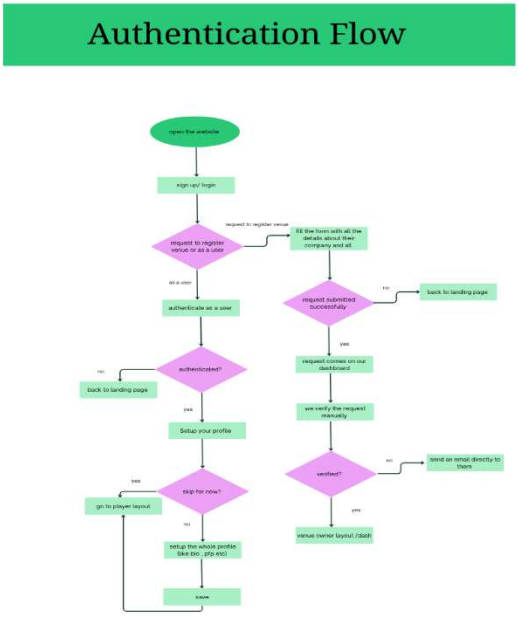


Figure 1: Authentication Flow — role selection, venue-owner request verification, and player profile setup.

4.1.2 Player Role — Booking of Venue with Payment Feature

Covers browsing venues and courts, choosing a timeslot, matching with other players, split payment, booking invitations, and checkout/payment verification.

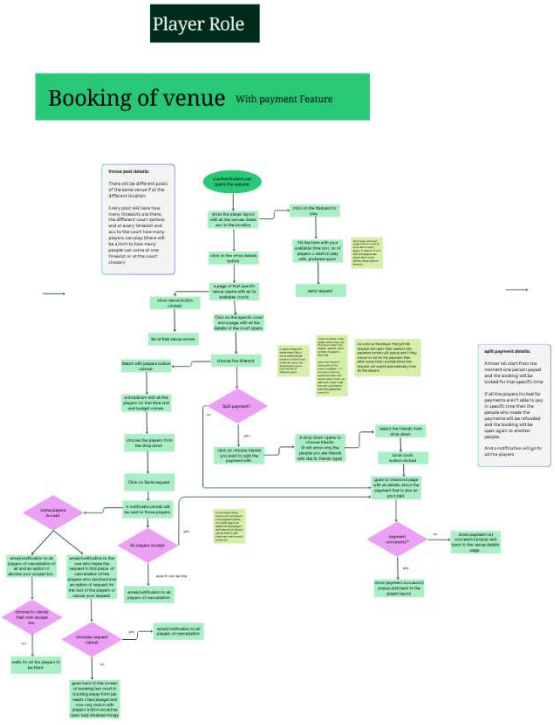


Figure 2: Player Role — venue and court booking flow with the split-payment feature.

4.1.3 Venue Owner Role — Dashboard

Covers the venue owner dashboard: viewing bookings, adding a venue, editing venues and courts (with the booking-conflict restriction), and deactivating a venue

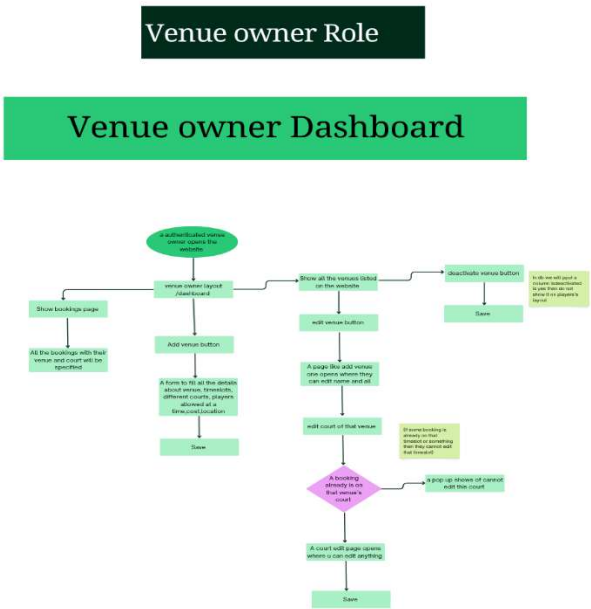


Figure 3: Venue Owner Role — dashboard for managing venues, courts, and bookings.

4.1.4 Player Role — Go Live Feature

Covers the social “go live” feature, which is visible only to a player's friends, similar to Instagram Live.



Figure 4: Player Role — Go Live feature and friend notification.

4.1.5 Tournament Section

Covers both sides of the tournament workflow: venue owners creating tournament packages, and players either booking a package directly or submitting a custom request for proposal (RFP).

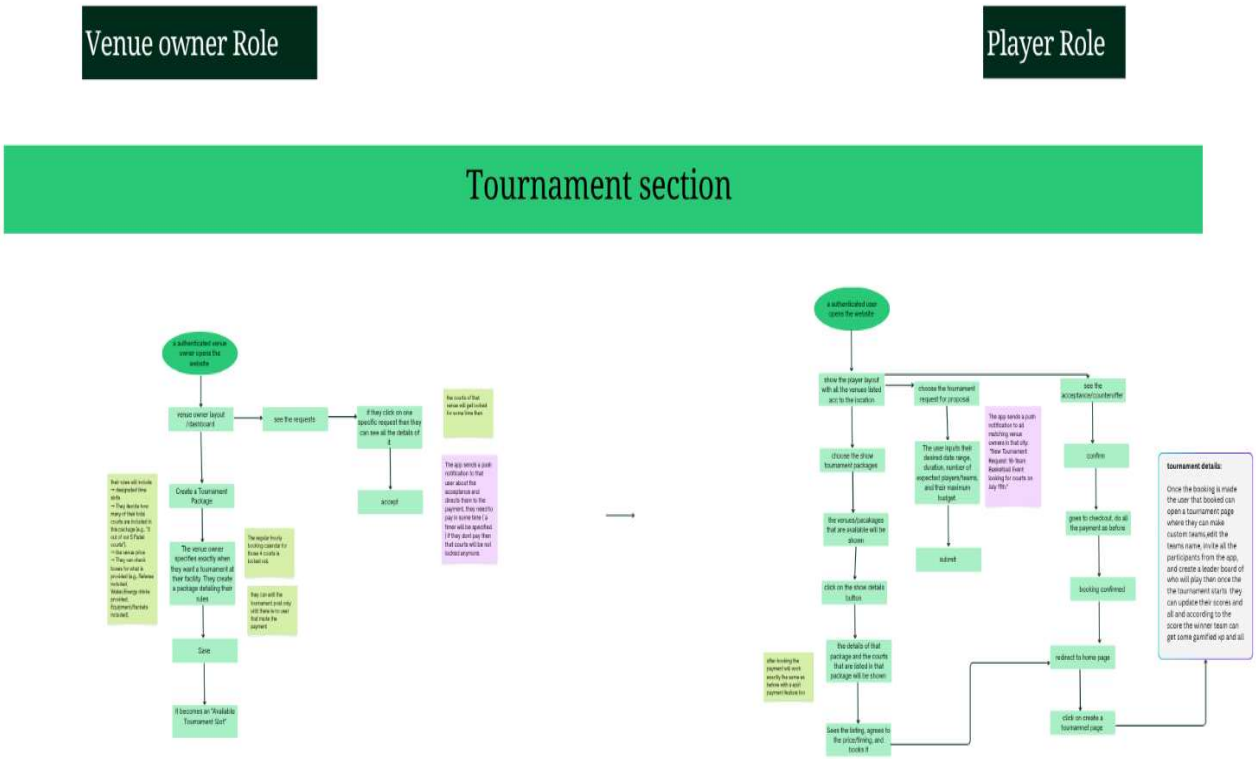


Figure 5: Tournament Section — Venue Owner package creation and Player booking / request-for-proposal flow.

4.2 Use Case Diagram

The use case diagram below consolidates the actors and actions identified across all four process flows: the Player, the Venue Owner, and the Admin who manually verifies venue owner registration requests.

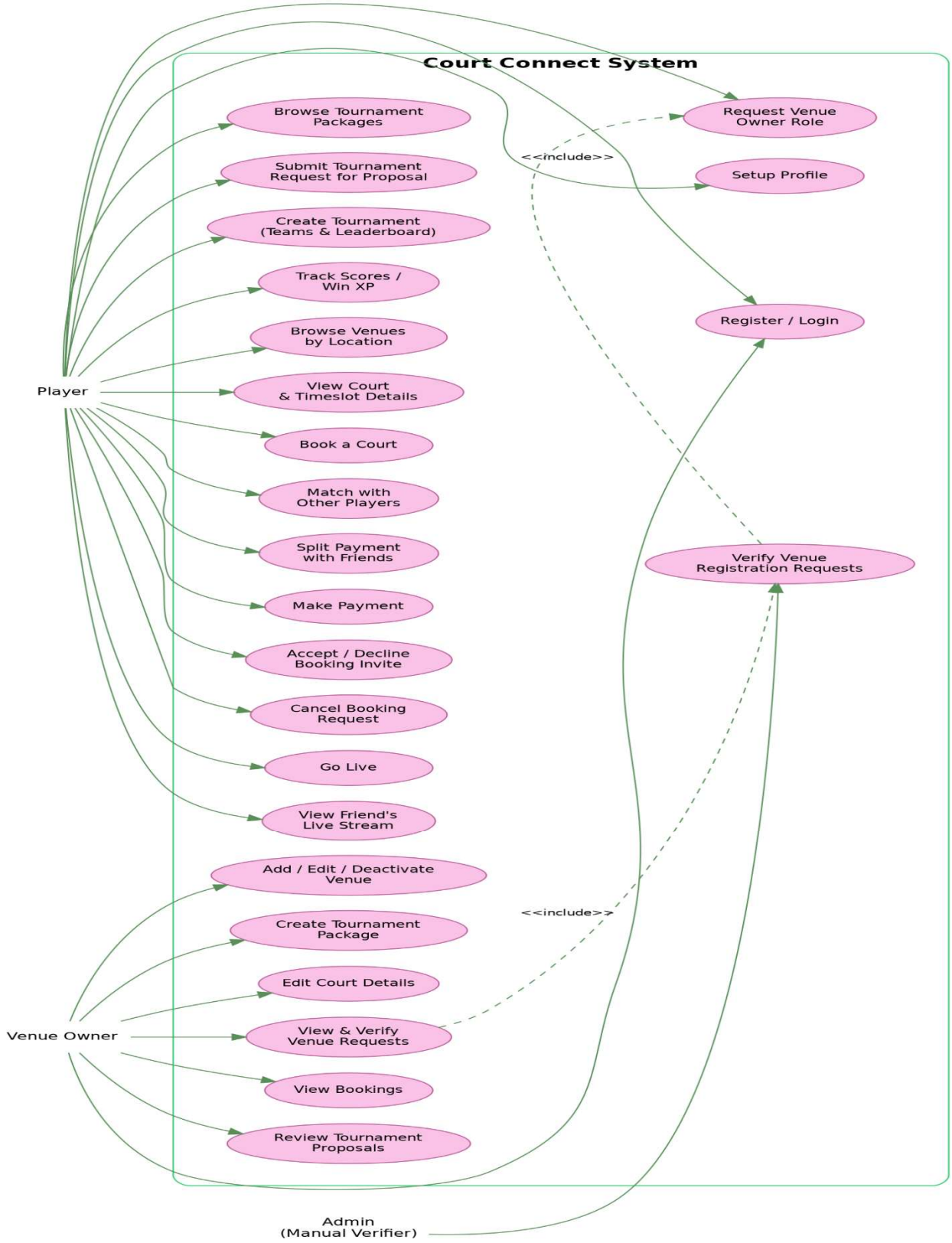


Figure 6: Use Case Diagram for Court Connect, showing Player, Venue Owner, and Admin actors.

4.3 Class Diagram

The class diagram models the core entities identified in the process flows — users, venues, courts, timeslots, bookings and their participants, payments, live sessions, and the tournament package/request/team hierarchy — along with their key attributes, operations, and relationships.

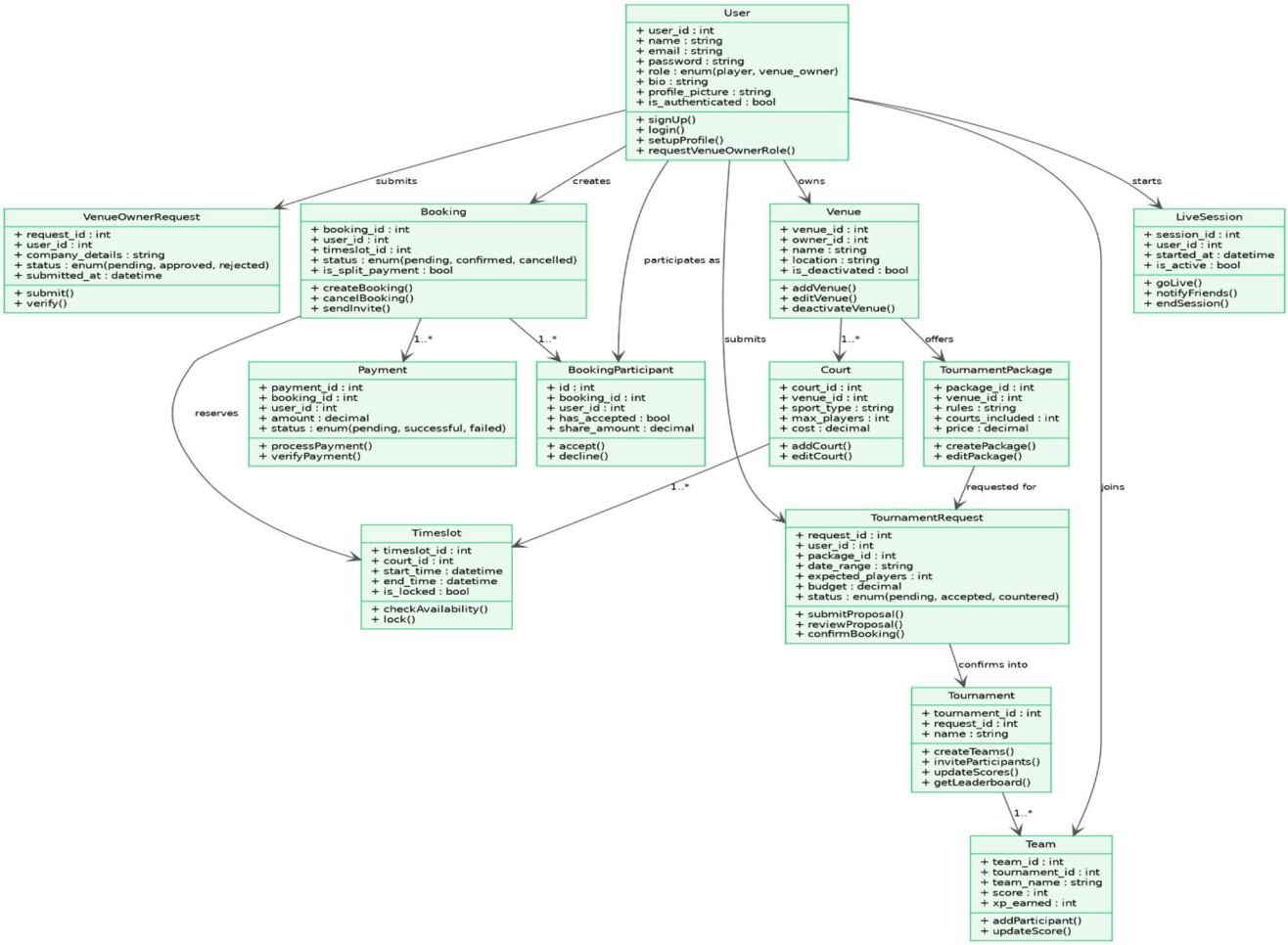


Figure 7: Class Diagram for Court Connect.

4.4 Entity-Relationship Diagram (ERD)

The ERD below shows the underlying database entities that support venue and court management, bookings, payments, and the tournament system, along with their key attributes and relationships.

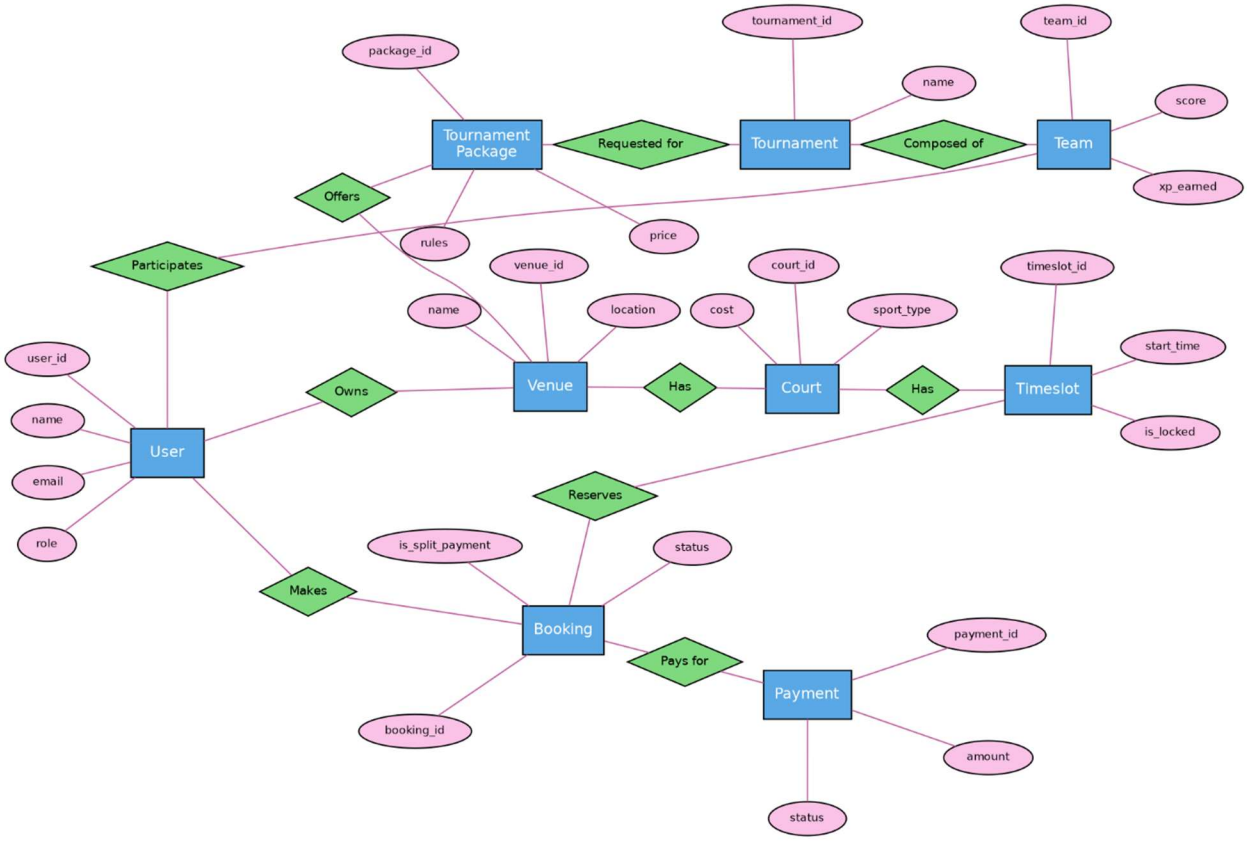


Figure 8: Entity-Relationship Diagram (ERD) for Court Connect.

5. Project Features

Player Features

- **Registration and Login:** Players sign up or log in and can optionally request to be onboarded as a Venue Owner instead.
- **Profile Setup:** Players can set up a bio and profile picture immediately, or skip and complete it later from their layout.
- **Browse Venues by Location:** Venues are listed according to the player's location, each showing all its available courts.
- **Court & Timeslot Selection:** Players view a court's available timeslots along with pricing and the number of players allowed.
- **Solo Booking:** Players can submit a request-to-play form specifying their desired timeslot, number of players, and preferred sport.
- **Match with Players & Split Payment:** Players can invite others already booked in a slot, or select friends from a dropdown, to split the cost of a booking automatically.

- **Booking Invitations:** Invited players receive a notification/email and can accept or decline; the requester is notified of the outcome.
- **Timed Checkout:** A countdown timer governs the payment window for a split-payment booking; unpaid bookings expire automatically and release the timeslot.
- **Go Live:** Players can broadcast a live stream from their profile, visible only to friends, with an automatic notification sent to them.
- **Browse & Book Tournaments:** Players can browse a venue's available tournament packages and book one directly.
- **Tournament Request for Proposal (RFP):** Players can submit a custom tournament request (dates, expected players, budget) that is broadcast to matching venue owners in their city.
- **Create & Manage a Tournament:** Once a tournament booking is confirmed, players can create custom teams, invite participants, and open the tournament page.
- **Live Scoring & Gamified XP:** Participants update scores during the tournament; the winning team is recognized on a leaderboard and rewarded with experience points (XP).

Venue Owner Features

- **Verified Onboarding:** Prospective venue owners submit their company and venue details for manual verification before gaining dashboard access.
- **Bookings Overview:** Venue owners can view all bookings made against their venue's courts from a dedicated bookings page.
- **Add Venue:** Venue owners can add a new venue by specifying courts, timeslots, players allowed per slot, cost, and location.
- **Edit & Deactivate Venue:** Venue owners can edit venue details or deactivate a listing so it stops appearing to players, without losing historical data.
- **Edit Courts with Conflict Protection:** Venue owners can edit a court's details, but the system blocks edits to a timeslot that already has an active booking.
- **Create Tournament Packages:** Venue owners can define tournament packages with designated courts, time slots, rules, and pricing.
- **Review Tournament Requests:** Venue owners can view and respond to player-submitted tournament requests for proposal, including counter-offers.

Platform / Admin Features

- **Manual Venue Verification:** An admin reviews and approves or rejects venue-owner registration requests before a venue can go live on the platform.
- **Automatic Slot Release:** The platform automatically expires unpaid split-payment bookings and releases the corresponding timeslot.
- **Notification System:** Email and push notifications keep players and venue owners informed of booking invites, acceptances, declines, tournament requests, and go-live alerts.

6. Technology Stack

6.1 Frontend

The frontend of Court Connect is built using **Next.js**, a React-based framework that provides server-side rendering and file-based routing. It powers the player-facing pages (venue browsing, court booking, split-payment flow, go-live feature) as well as the venue owner dashboard, delivering a fast, SEO-friendly, and responsive interface across desktop and mobile.

6.2 Backend

The backend is built using **NestJS**, a progressive Node.js framework that structures the application into modules, controllers, and services. NestJS handles the REST API layer, including authentication, venue and court management, booking and split-payment logic, tournament requests, and notification dispatch.

6.3 Database

PostgreSQL is used as the relational database, managed through **Supabase**. Supabase provides the hosted PostgreSQL instance along with built-in authentication, real-time subscriptions, and storage, which support features such as live booking status updates and profile picture uploads.

7. Project Outcome

This section presents the final, working interface of Court Connect through screenshots of the deployed application, covering the player-facing website, the venue owner dashboard, and the tournament module.



Figure 9: Home page hero section of our project.

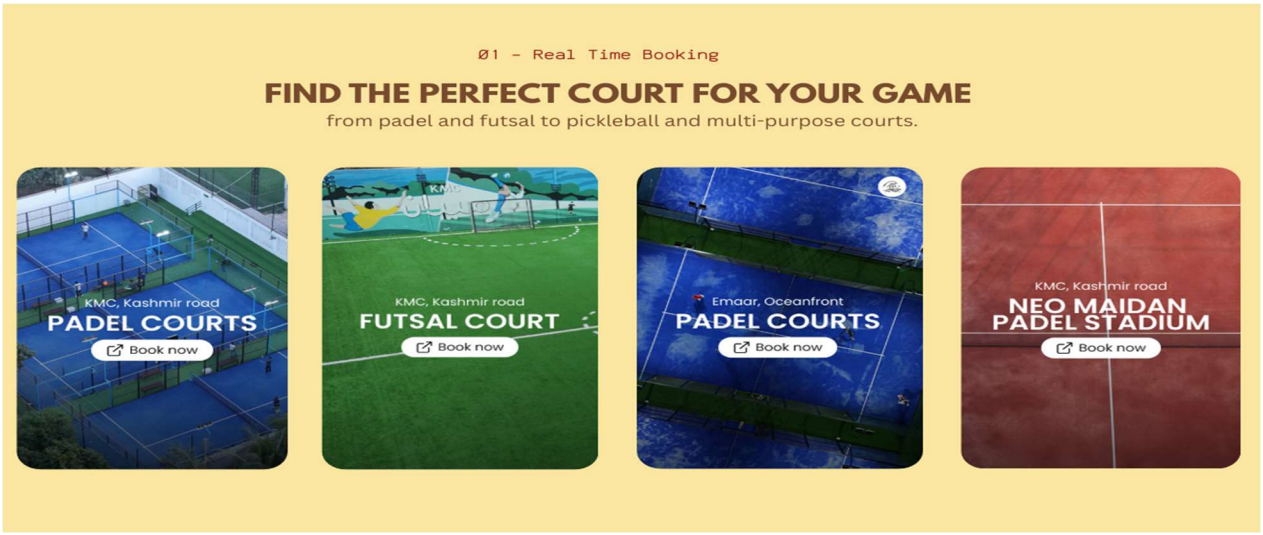


Figure 10: Shows how real time booking looks like.

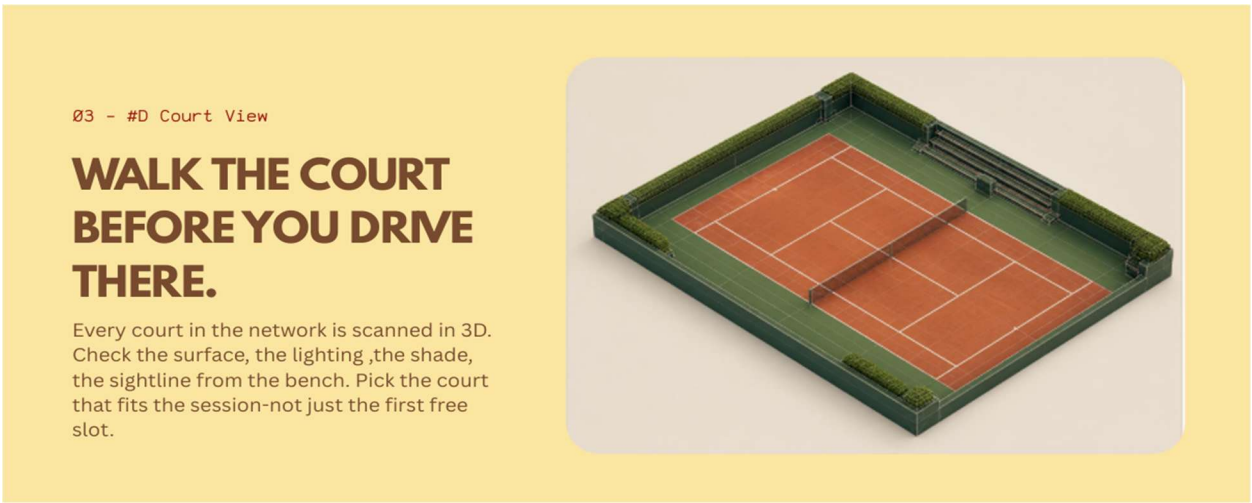


Figure 11: User can view the court before booking.



Figure 12: User Venue booking page.

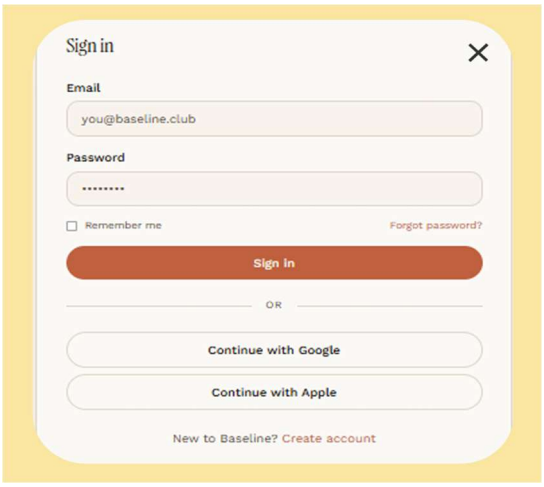


Figure 13: Sign in requirements.

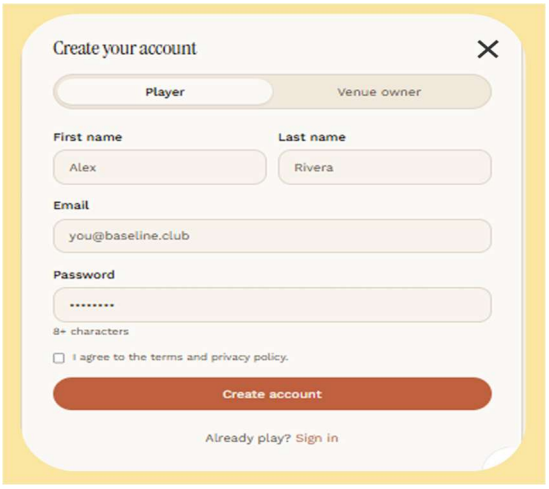


Figure 14: More requirements.

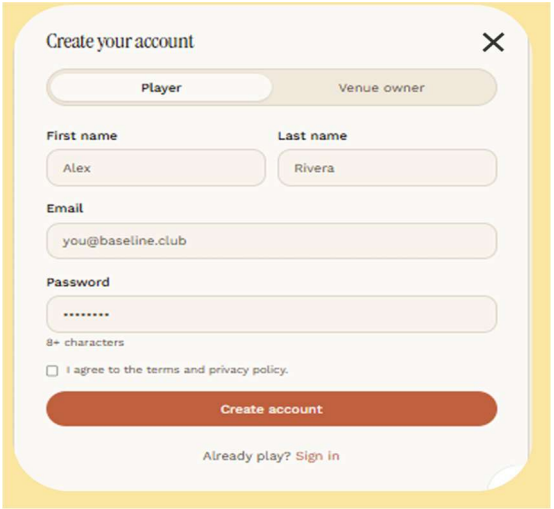


Figure 15: Create user account.

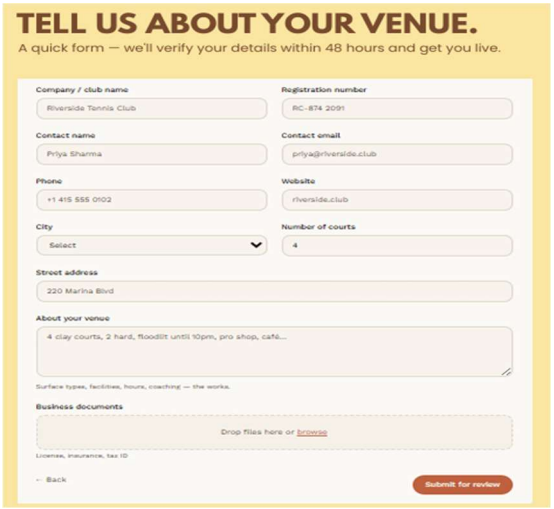


Figure 16: A quick verification form.

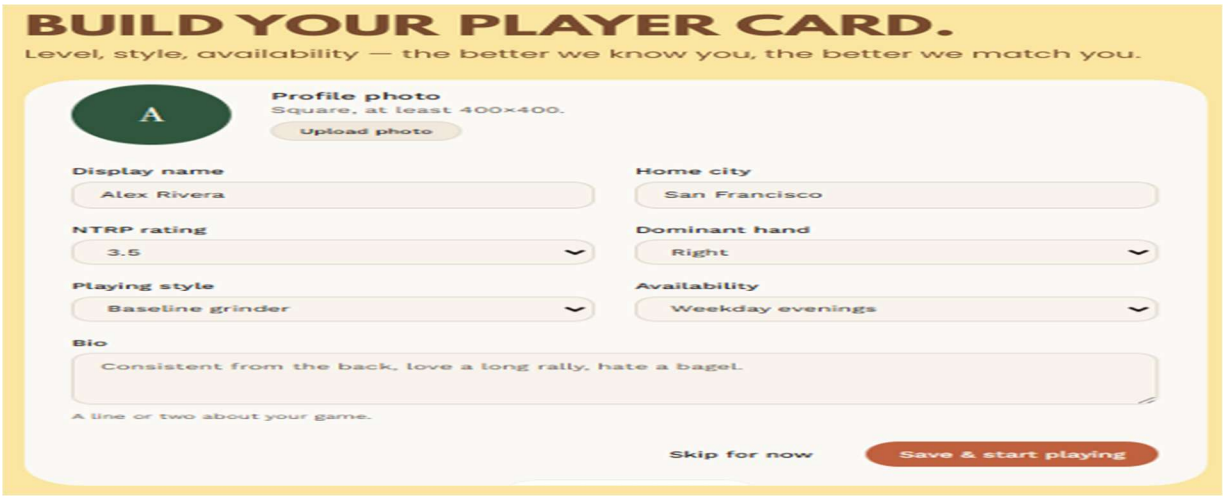


Figure 17: Player card so matching with others could be easy.

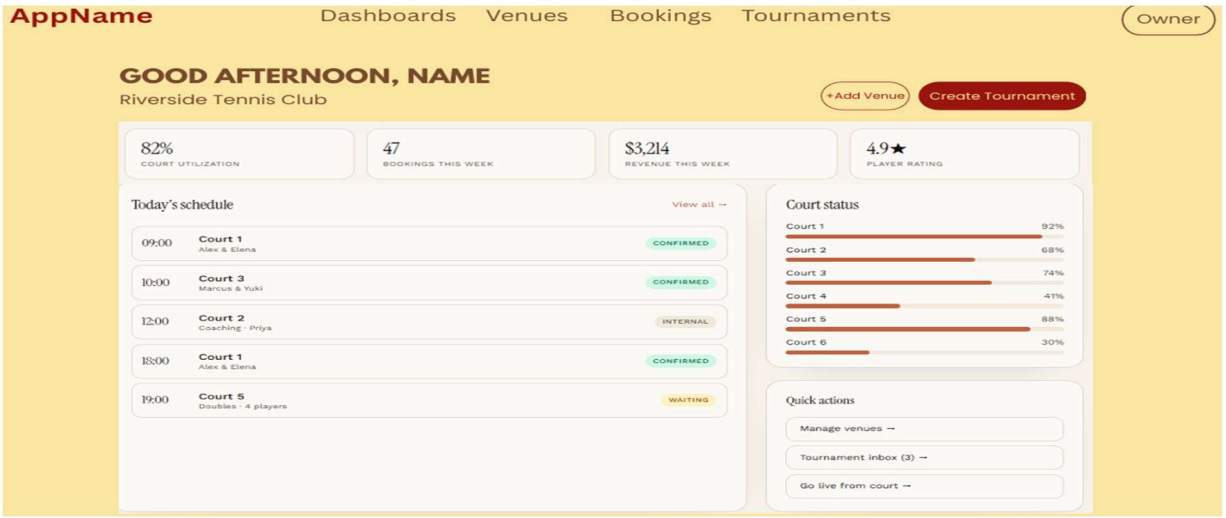


Figure 18: Venue Owners dashboard.

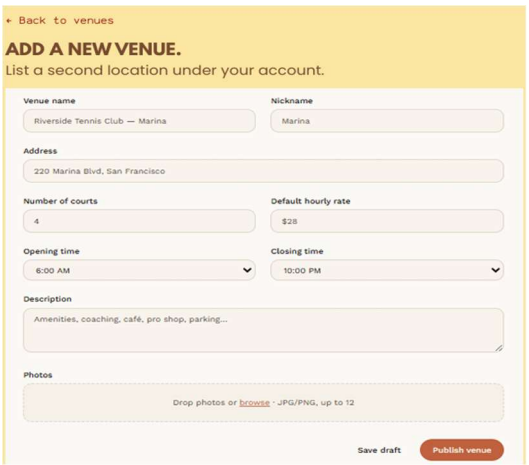


Figure 19: Business owner to register venue.

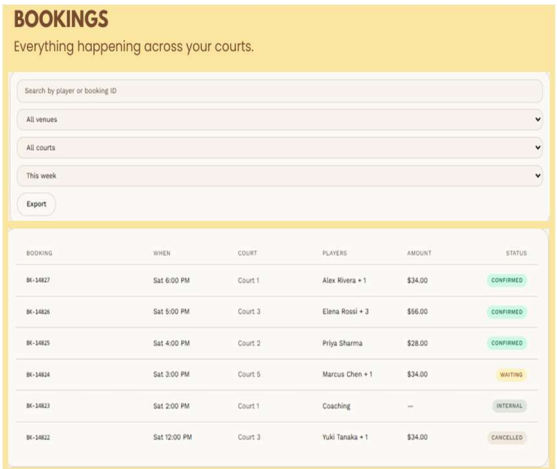


Figure 20: Owners can watch around court.

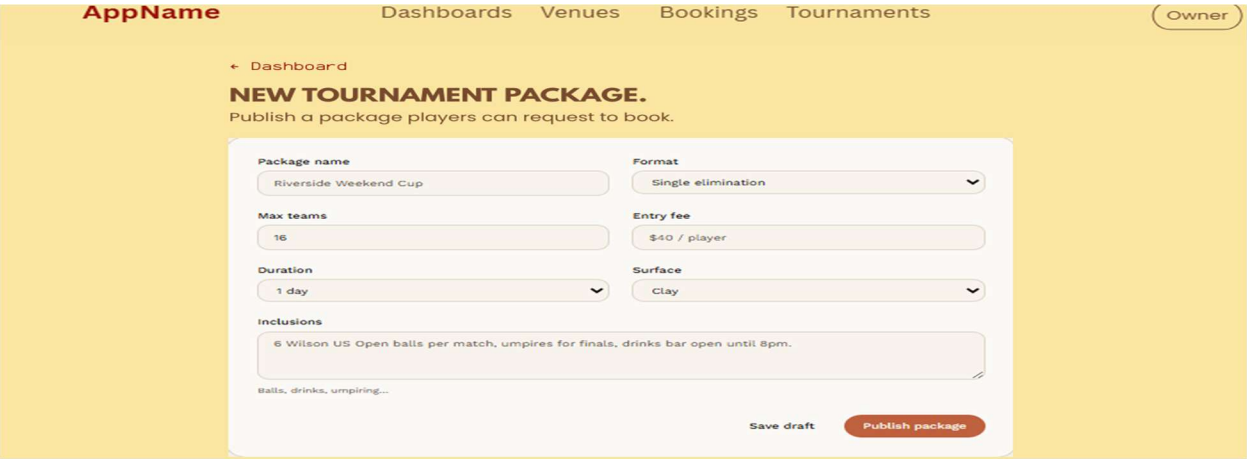


Figure 21: Court owner publishes tournament package.

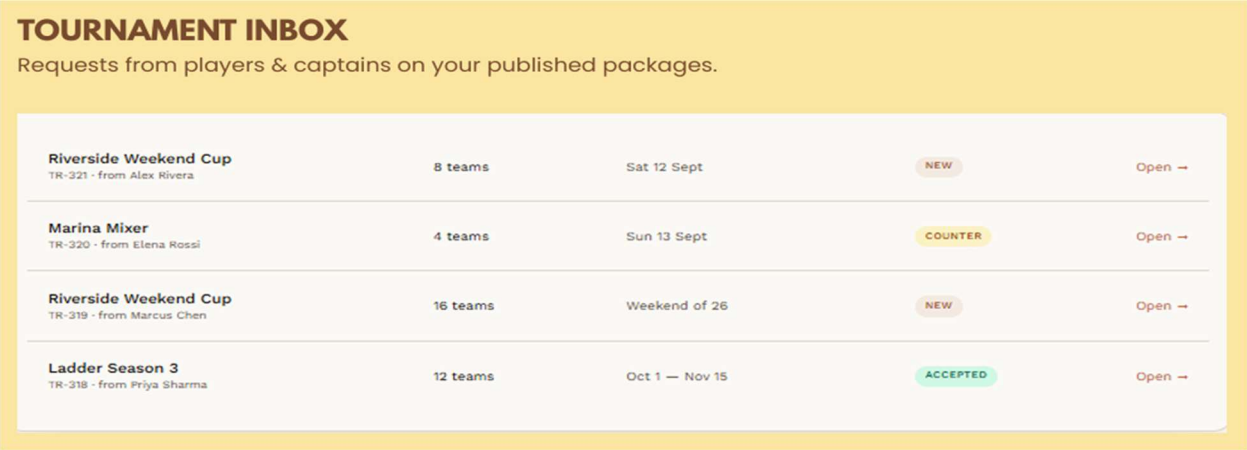


Figure 22: Response of players toward the tournament package.

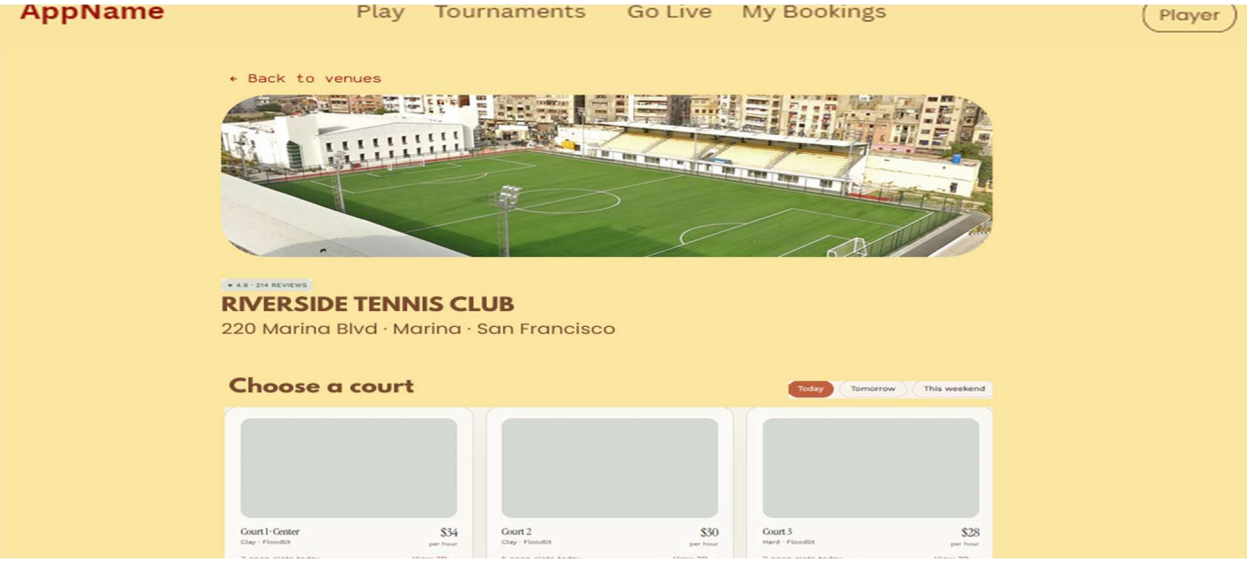


Figure: 23: Players court booking page.

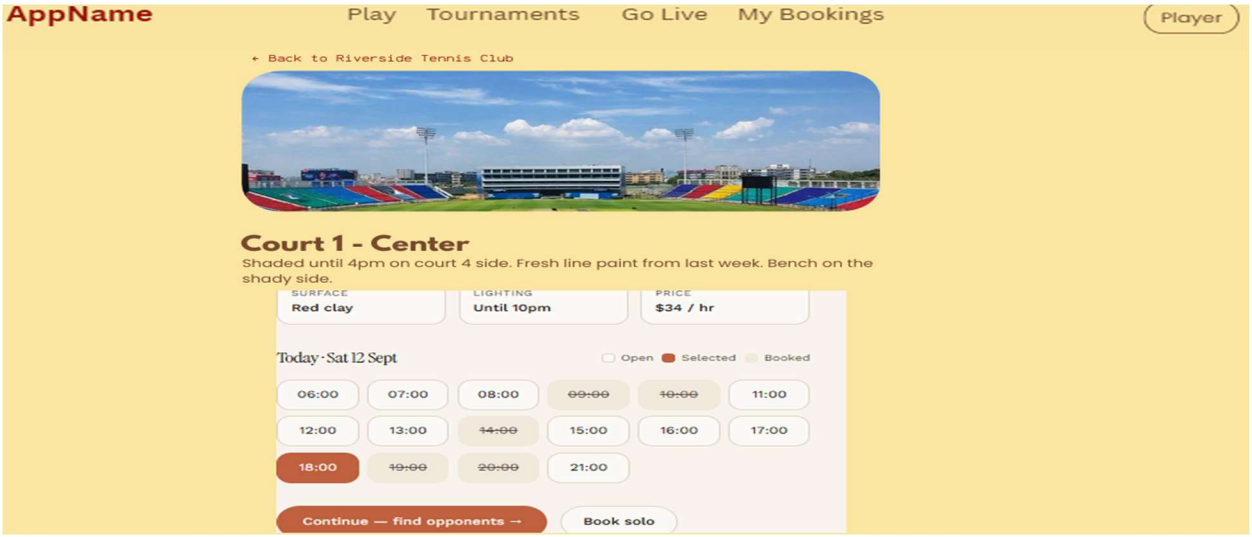


Figure 24: Selecting time slot user wants to book.

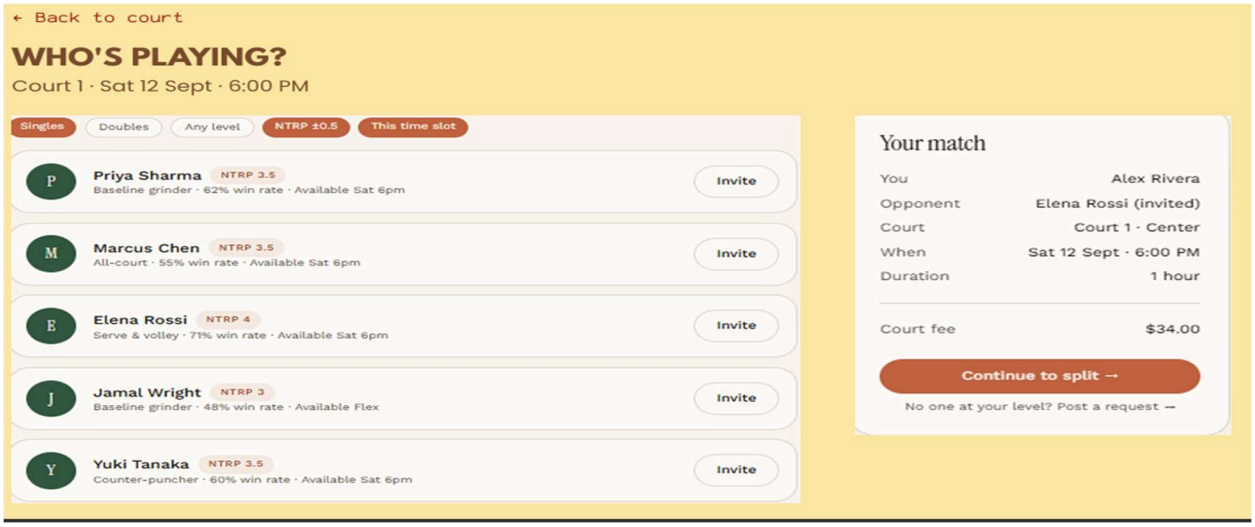


Figure 25: Player choose his/her friends or match with anyone to play send them invite.

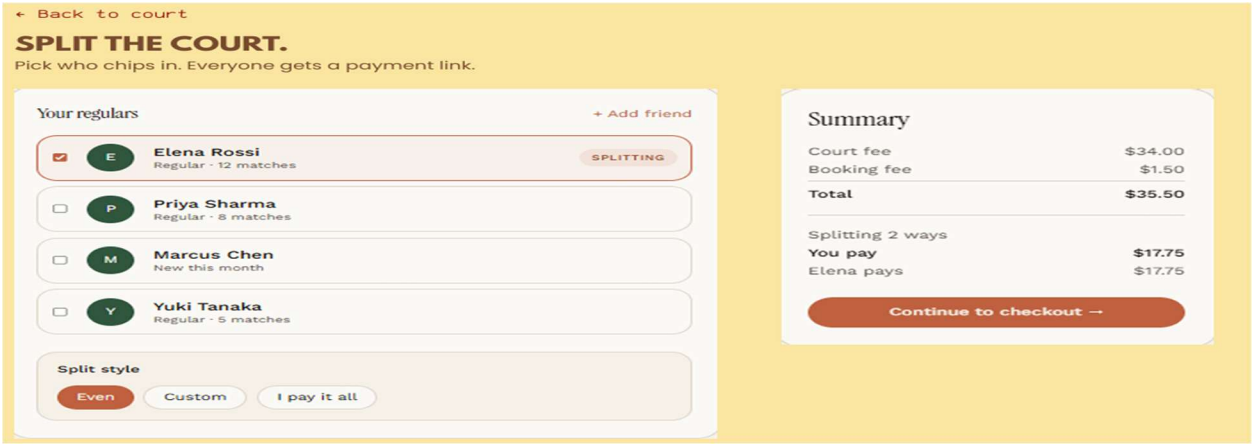


Figure 26: Split payment feature where everyone gets payment link.

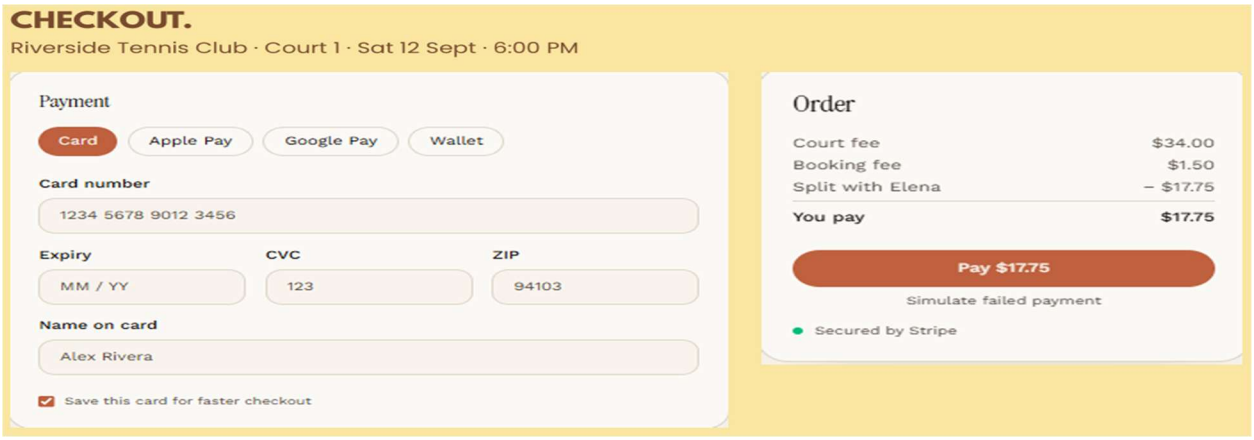


Figure 27: Checkout Phase.

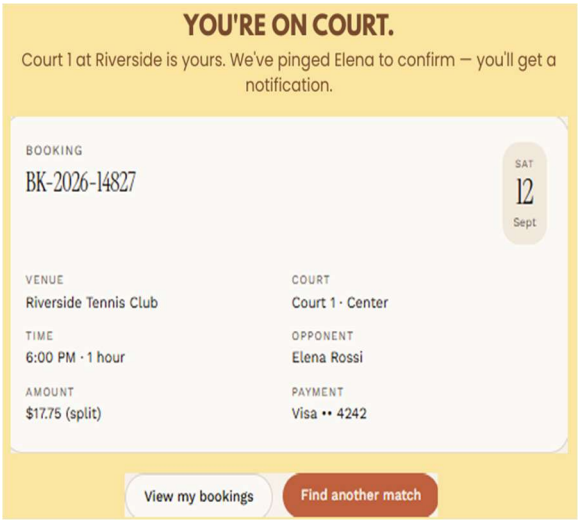


Fig 28: Payment done! You're in

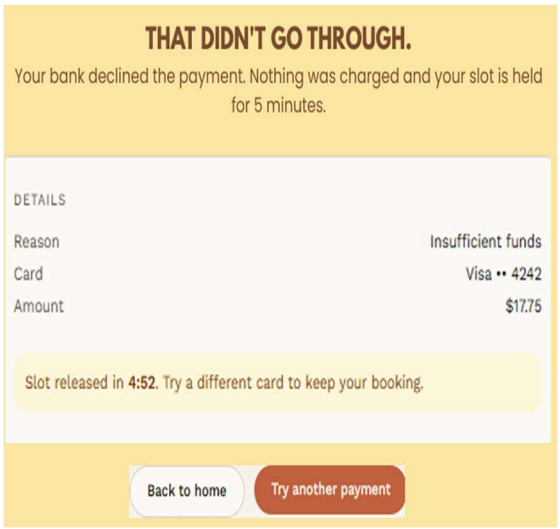


Fig 29: Payment lost. Slot is held for 5 mins

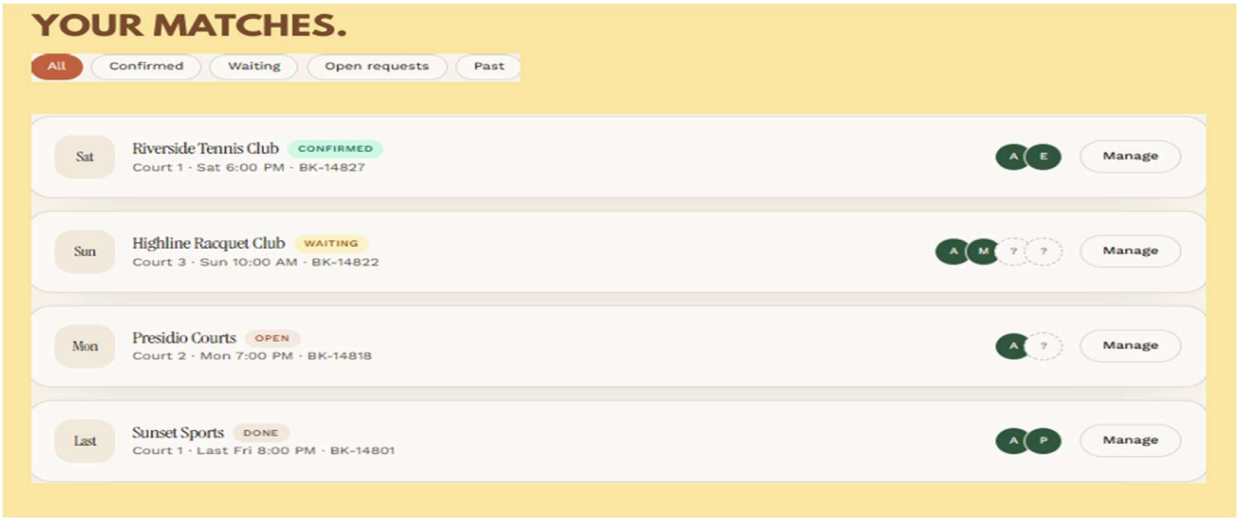


Figure 30: Player can see potential matches.

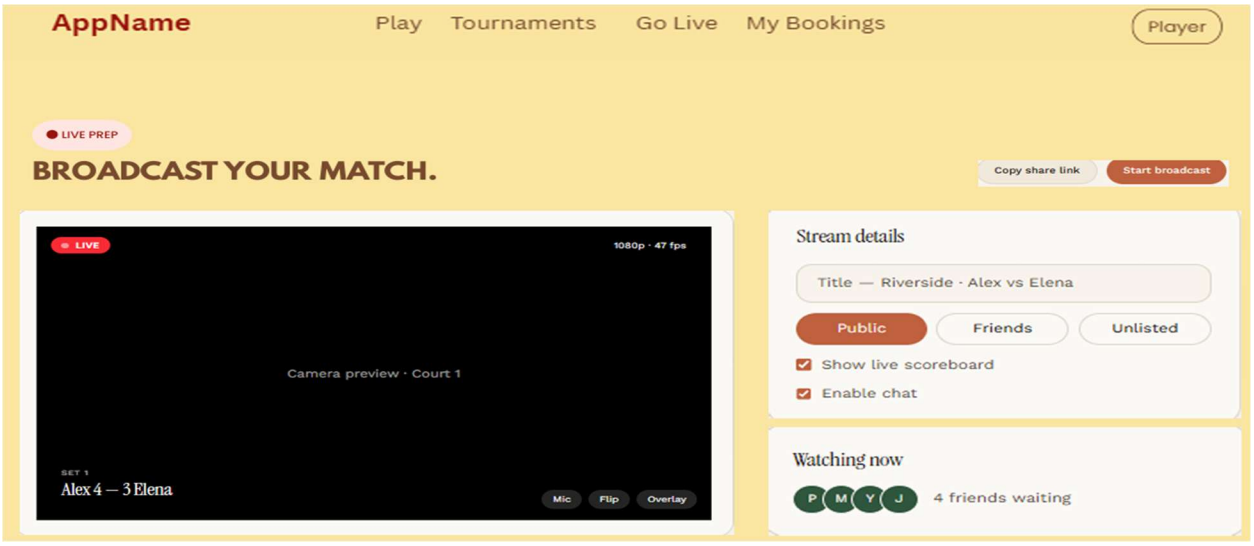


Figure 31: Broadcast / Livestreaming page.

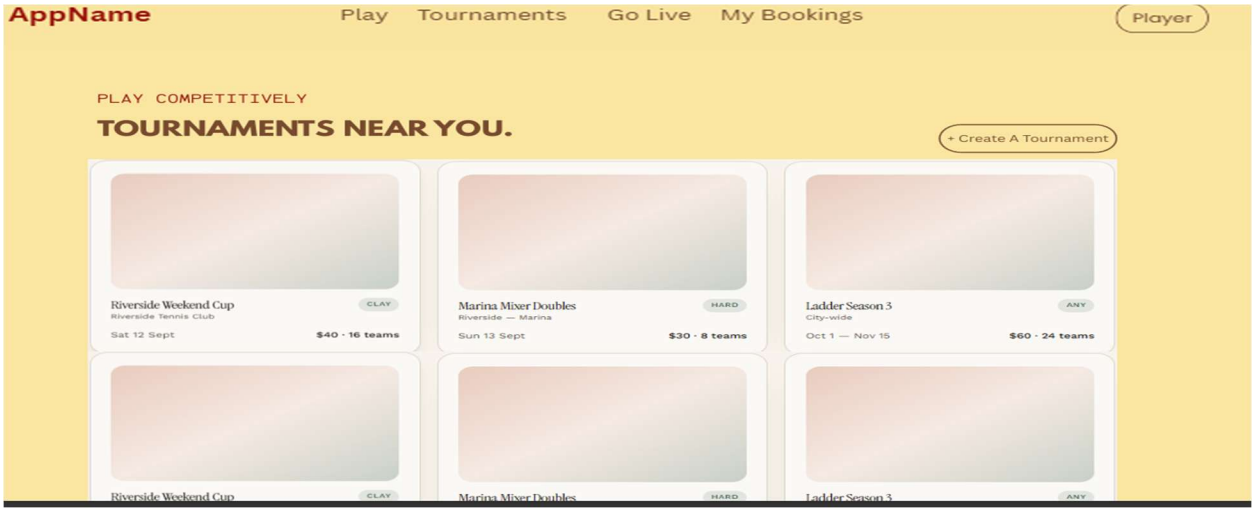


Figure 32: Players can check courts hosting tournaments near them.

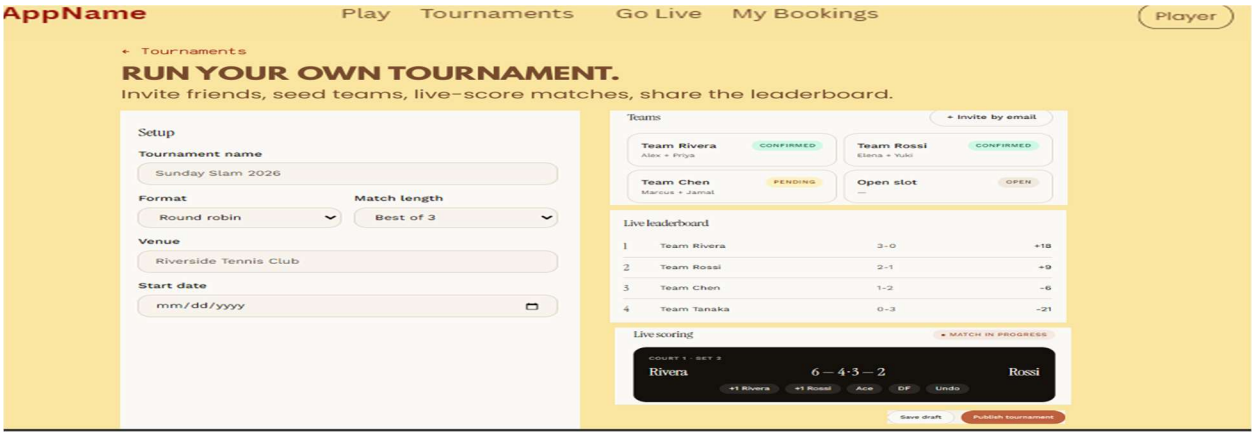


Figure 33: Players can customize their own tournament.

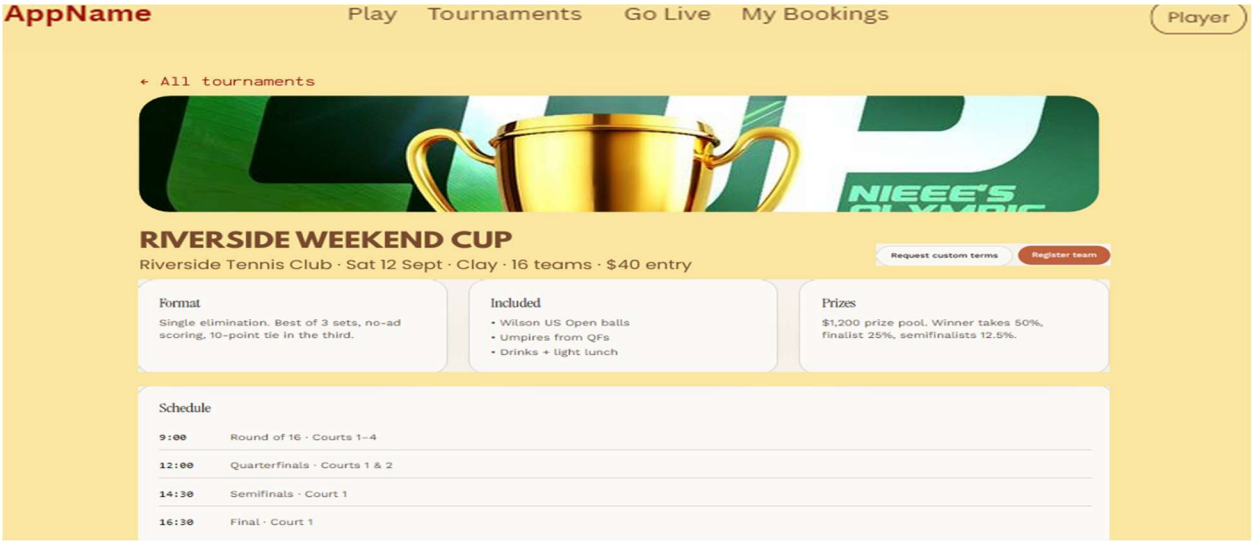


Figure 34: Tournament details, can opt for custom requests too.

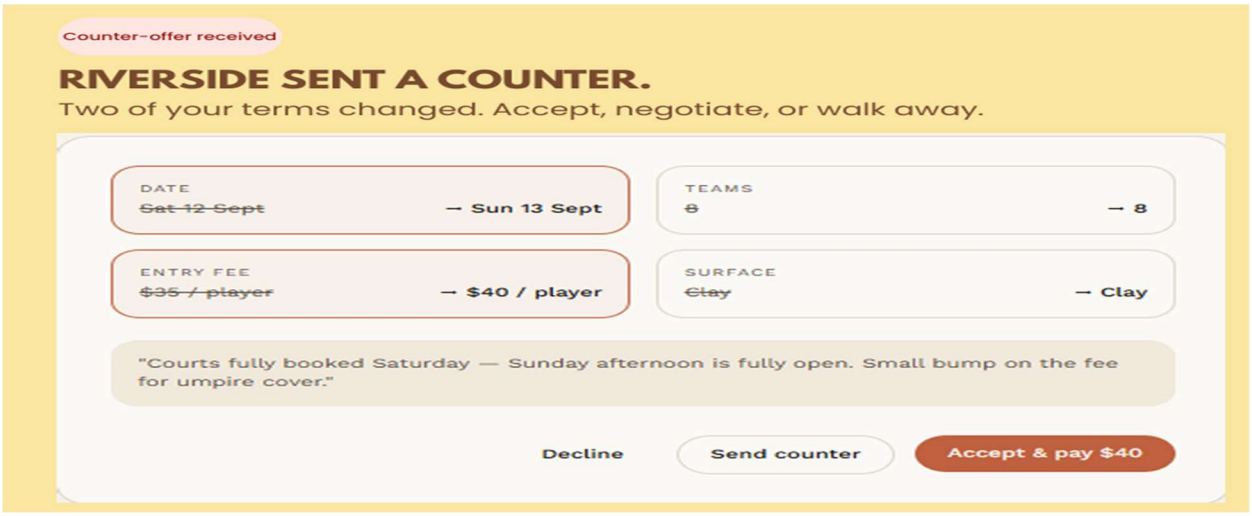


Figure 35: Counter offer received offered by venue owner.



Figure 36: Players propose their requests and sends a proposal.